

come from all over the globe, and people are committed to improvement and are not afraid to experiment. And the best part is that even if the development is dropped, nobody stops you from developing the software further and creating a solution in your very own style.

Each EDA tool will have several sub-tools for different functions like schematic entry, PCB design, simulation, Gerber view, etc. Let us review these tools one by one and understand the basic parameters and functionalities for evaluating them.

Schematic capture

Schematic capture or schematic entry is the very first step in the design cycle of an electronic system. This is normally not done all at once but the overall design is broken into sub-modules. The schematic of these smaller portions are entered in the schematic capture tool using the in-built symbols which are wired together to represent the design. The design of these sub-modules is verified using simulation programs. Similarly, other sub-modules are entered and verified for functionality one after another. These sub-modules are then integrated with one another and again the simulation helps to check if they work as expected, even together. In this manner, the complete design is laid out using the schematic capture tool where simulation and schematic entry go hand in hand.

Now that we have seen how important a schematic capture tool is, look at some aspects that might help you in selecting one.

User friendliness. When you are going to spend so much time with this tool for all your designs, it is important that it is user-friendly, organised, simple and yet powerful. All basic functions such as selecting components from the library, placement, attribute entry, wiring, editing, navigating the design, zoom, etc, should be straightforward and intuitive. These are very basic functions but make a very big difference to a designer. For example, some tools have a better library search

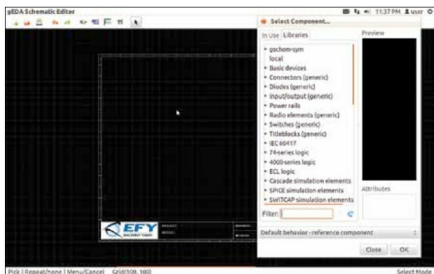


Fig.3 Schematic capture tool 'gschem'

system than others, enabling you to find the parts very easily with just some common sense, while with others you have to put in a lot of effort to find the correct part before you can include it in a schematic.

In-built library. No library can have all the available electronic parts because every day there are new entrants. But the more accomplished the symbol library is, the less time you will waste making new symbols. So do check the libraries thoroughly and compare it with other tools to have an idea. There are various open source tools that have communities contributing to these libraries. So if you do not find a part in the in-built library, you can look in the contributed libraries and there is a good chance that you will find your required part. Such a feature is a definite plus point while selecting an EDA tool.

Symbol creation. No matter how accomplished the in-built symbol libraries are, you will always have to make new symbols for your designs. Do make some symbols yourself while evaluating a tool to see if it is really quick to make and include them in the existing libraries. Some tools have a very complicated procedure for making new symbols and including them in your design, which results in designers being reluctant to make new symbols and they try to work around

the problem with the existing ones.

Hierarchical designs. This is a must. When the design gets complex, it is better to represent it as a hierarchical design. This makes the representation simple, traceable, and easy to navigate and review. In schematics with a lot of repetition, this feature can save you labour. For example, if you have 10 channels of some circuitry, you can build it once and replicate it using hierarchy. Second, when multiple persons are working on a project, it enforces clean boundaries. Because all high-level connectivity is via ports, there is no risk of accidental connectivity via off-page connectors. And yet, within each subsection of the design, you can still use off-page connection if you like. Also check the page limitation, if any, for multi-sheet designs.

ERC (electrical rule check). The ERC is used to test schematics for electrical errors such as unconnected inputs, shorted outputs, correct power and ground connections. When you are working on a schematic, there will be some human errors that can lead to tears at a later stage of development. This feature can catch such silly mistakes in time to save you from a lot of suffering.

Smart features. Smart features such as auto-number, replace, slot, etc, are highly useful. For example, if there are 20 resistors in your design and